Introduction to Human Sciences

Philosophy Module — Endsem Synthesis

"The Region of Liberating Doubt"

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How to use this document. The four lecture-sections below reconstruct the argumentative arc of the module: from the very meaning of *philosophical inquiry* (§1), through the formal distinctions of *logic* (§2), the engine of *deductive vs. inductive inference* (§3), and finally the catalogue of *informal fallacies* (§4). Each section contains: a thematic overview, formal argument reconstructions (numbered Premise / Conclusion form), a philosophy lexicon, profiles of the thinkers invoked in class, and a list of counter-arguments raised in the transcripts. §5 and §6 consolidate the lexicon and exam-style preparation.

Contents

1 April 9 — What is Philosophy? The Nature of Inquiry

Thematic Overview

The opening lecture stages a question whose answer is itself contested: *what distinguishes a philosophical question from a scientific one?* Prof. Jayanti's thesis is that philosophy is the discipline that confronts **normative** and **conceptual** questions which empirical science cannot, by its very method, resolve. Far from being "mere opinion," philosophy is **grounded in logic** and proceeds by *argument*—the giving of reasons whose coherence one can publicly test. The lecture closes by quoting Russell: philosophy's value lies not in delivering certainty but in transporting us into "*the region of liberating doubt*."

1.1 The Meaning of Philosophy

The word "philosophy" derives from the Greek *philo* (love) and *sophia* (wisdom): literally, *the love of wisdom*. It is the oldest surviving intellectual discipline in the Western tradition. Almost every subject we study today—biology, physics, psychology, economics, AI—branched off from philosophy at some point in history, as certain questions got answered satisfactorily and became their own fields. What philosophy does is *stick with the unanswered questions*.

A common misconception is that philosophy is esoteric or impractical. Stich and Donaldson (assigned reading for the module) show that philosophical questions arise in everyday life constantly. Consider whether it is okay to eat meat: the question leads immediately into animal consciousness, the relationship between pain and the nervous system, the existence of God, and the reliability of scripture—all philosophical questions. Further examples of philosophical questions everyone confronts sooner or later:

- Is it possible to travel backward in time?
- Do we have free will?
- What is the difference between knowledge and opinion?
- Could a digital computer be conscious?
- When, if ever, is it okay to lie?

1.2 The Normative / Descriptive Distinction

Core Argument 1: Why Philosophy Cannot Be Reduced to Science

P1. Scientific (descriptive) questions concern *what is the case*; they are answerable by observation, experiment, and statistical method.

P2. Philosophical (normative) questions concern *what ought to be the case*; they ask for *justification*, not *causal explanation*.

P3. No purely descriptive set of premises logically entails a normative conclusion without a smuggled value-laden premise (the *is–ought gap*, after Hume).

P4. Therefore, science alone cannot resolve normative questions: ethical, political, aesthetic, and conceptual disputes irreducibly require philosophical work.

∴ **C.** Philosophy is a distinct, indispensable discipline; the reduction "science has replaced philosophy" (Hawking) commits a category-mistake.

Worked illustration (capital punishment). Even granted complete empirical data on deterrence rates, the inference "crime fell, *therefore* we should retain capital punishment" presupposes the unstated normative premise: "one ought to choose the policy that minimises crime regardless of moral side-constraints." This is the **hidden premise** that makes the inference go through—and makes it *philosophical*.

1.3 What Concerns Philosophy?

According to the slides, philosophy concerns four interlocking question-types:

(i) Questions science cannot answer (e.g. does free will exist?).

(ii) **Justification** of claims (epistemic warrant, not causation).

(iii) **Analysis of concepts** (knowledge, justice, intelligence).

(iv) **Normative** questions (what ought we to do/believe?).

1.4 Domains of Philosophy — A Lexicon

Philosophy Lexicon I — The Domains

Logic
The study of *right reasoning*: which inferences are valid?

Epistemology
The study of *knowledge*: how is knowledge possible? What are its conditions?

Ontology / Metaphysics
What *exists*? What is the relation between mind and world?

Ethics
What is right or wrong, good or bad?

Axiology
The general theory of *value*.

Aesthetics
What is *beauty*? Is it objective or convention-bound?

Philosophy of X
The reflective examination of any discipline's foundational assumptions, where $X \in \{\text{law, science, mathematics, AI, sport, history, } \ldots\}$.

1.5 The Philosophical Attitude: Socratic Irony & Liberating Doubt

Core Argument 2: Why Doubt is "Liberating" (Russell)

P1. Most beliefs we live by are inherited as "readymade truths" (custom, dogma, popular consensus, scientism).

P2. Without scrutiny, such beliefs masquerade as knowledge while remaining unjustified.

P3. Philosophical inquiry, by suspending these beliefs, suggests *many possibilities* the inheritor of dogma never sees.

P4. Hence philosophy "*enlarges our thoughts and frees them from the tyranny of custom*" (Russell, *The Problems of Philosophy*).

$\therefore$ **C.** The value of philosophy lies in **what we may be**, not in delivering settled answers about what *is*; it removes "the arrogant dogmatism of those who have never travelled into the region of liberating doubt."

Socratic Irony. The Delphic oracle named Socrates the wisest man in Athens. His response: "*All I know is that I know nothing.*" Wisdom begins in epistemic humility—a corrective, the lecture stresses, to the **Dunning–Kruger effect**: the less one knows, the more confidently one overestimates one's knowing.

1.6 Philosophy as the Root of Other Disciplines

- Newton was a *natural philosopher*; his magnum opus is *Mathematical Principles of Natural Philosophy*.
- Russell & Whitehead's *Principia Mathematica* sought a logical foundation for arithmetic (famously taking ~300 pages to prove $1 + 1 = 2$).
- Turing's "*Computing Machinery and Intelligence*" was published in the philosophy journal *Mind*; it founded AI by proposing the *imitation game* as a definition of intelligence.
- Searle's **Chinese Room** is the canonical philosophical critique of that very definition: syntax (symbol manipulation) alone cannot suffice for semantics (meaning), so a Turing-passing program would not thereby be minded.

1.7 Thinker Profiles (Lecture 1)

Thinker Profile: Socrates (c. 470–399 BCE)

Stance: Knowledge begins in the recognition of one's own ignorance. Philosophy is a *practice* of dialectical questioning, not a body of doctrine.
Cited in class: "All I know is that I know nothing"; "*the unexamined life is not worth living.*"
Key concept: *Socratic irony.*

Thinker Profile: Plato (c. 428–348 BCE)

Stance: Knowledge is *justified true belief* (the classical "JTB" analysis). Truth is not what is empirically observed but what survives rational scrutiny.
Cited in class: *The Republic* (foundational text of political theory); definition of knowledge as JTB; democracy as risk of majoritarian tyranny.

Thinker Profile: Bertrand Russell (1872–1970)

Stance: Philosophy's value lies not in answers but in liberating us from dogma; logic is the bedrock of rational thought.
Cited in class: *The Problems of Philosophy* (esp. "The Value of Philosophy"); *Principia Mathematica* (with Whitehead); the Russell–Coppleston debate on God's existence (recurring exemplar).
Key concept: "*the region of liberating doubt.*"

Thinker Profile: Karl Popper (1902–1994)

Stance: Science progresses by *falsification*, not verification: no theory is ever proved; the best we say is that it has not yet been refuted.
Cited in class: The *hypothetico-deductive method*; debate with the logical positivists.

Thinker Profile: Hilary Putnam (1926–2016)

Stance: Even those who scorn philosophy already practise it—badly. Hence: "*We need philosophy because bad philosophy is all around.*"

Thinker Profile: Immanuel Kant (1724–1804)

Stance, as invoked in class: Enlightenment is the "*Sapere aude*"—dare to think for yourself: setting aside ecclesial and popular authority and building one's beliefs from one's own reason.

1.8 Potential Critiques Raised in the Transcripts

Objection / Critique: Stephen Hawking's Eliminativism

"Philosophy is dead; science has taken over."

Reply (Jayanti). Hawking's claim is itself a philosophical claim: it makes a normative assertion about the proper sources of knowledge, which cannot be empirically verified. Moreover, every science rests on foundational assumptions (causation, induction, the uniformity of nature) whose *justification* is philosophical, not scientific—hence the active field of **philosophy of science**.

Objection / Critique: Naturalistic / Evolutionary Reductions of Ethics

A reductionist (e.g. Dawkins, *The Selfish Gene*) might argue that "humans are evolutionarily selfish, therefore selfishness is good/permissible."

Reply. The argument commits the *naturalistic fallacy*: it slides from a *descriptive* fact (we behave selfishly) to a *normative* prescription (we ought to be selfish), bridging the gap only by smuggling in an unstated value premise. The is–ought gap remains.

Objection / Critique: "Philosophy is Just Opinion"

Reply: a philosophical text is not an opinion but a *thesis*—a defended argument whose premises must give the reader a good reason to accept the conclusion. This is precisely why a Ph.D. is a *defence*; why the Russell–Coppleston debate begins not with rhetoric but with the joint articulation of definitions.

2 April 13 — Logic & Critical Thinking: Building Blocks

Thematic Overview

Having argued that philosophy is grounded in logic, the second lecture *introduces* that ground. Logic is here defined as the **study of rational argumentation**: a normative discipline whose job is to distinguish good arguments from bad. The lecture distinguishes *formal* from *informal* logic, isolates the *argument* as the basic unit of analysis, separates *sentence* from *proposition*, *truth* from *validity*, and prepares the ground (resumed in §3) for the formal study of inference.

2.1 What Logic Is—and Is Not

Defining Logic

Logic
: The study of rational argumentation; the *art of self-defence* against bad arguments, rhetoric, and propaganda.

Normative, not descriptive
: Logic asks how we *ought* to reason, **not** how we *do* reason. (The latter is the province of cognitive psychology, which catalogues confirmation bias, the Dunning–Kruger effect, etc.)

Formal logic
: Concerns the *form* or *structure* of arguments. Includes categorical propositions, syllogisms, propositional logic, predicate logic, and truth tables.

Informal logic
: Concerns the *content* or *meaning* of premises in everyday argument; centred on the identification of **fallacies** (treated in §4).

2.2 The Anatomy of an Argument

Core Definition: What an Argument Is

An **argument** is a structure in which *premises* are advanced to provide rational support for a *conclusion*. Its three building blocks:

Floor (Premises) The propositions assumed or asserted in support.

Roof (Conclusion) The proposition the argument seeks to establish.

Cement (Inference) The logical relation by which the premises are claimed to support the conclusion.

The whole structure rests on the **Laws of Logic** (the foundation).

2.2.1 Sentence vs. Proposition

Sentence vs. Proposition

Proposition
: The *meaning* expressed by a declarative sentence; bears a **truth-value** (true or false). Propositions are the building blocks of reasoning.

Sentence
: A *linguistic* entity. Two distinct sentences may express one proposition ("It is morning" / *Podduna ayindi* / *Subah ho gayi*); the same sentence in different contexts may express distinct propositions.

Non-proposition
A sentence that cannot be true or false: questions ("Will it rain?"), commands ("Get out!"), exclamations.

Worked case (the Liar's Paradox). "*This sentence is false.*" If true, it is false; if false, it is true. The sentence *looks* like a proposition but yields contradiction—a paradox that has driven entire research programmes in logic (Russell's type theory; Tarski's hierarchy of metalanguages).

2.3 Truth vs. Validity vs. Soundness

The Three-Way Distinction

Truth is a property of *propositions*: a proposition is true or false depending on whether it matches the way the world is.
Validity is a property of *arguments*: an argument is valid iff it is logically impossible for its premises to be true *and* its conclusion false.
Soundness = *Validity* + *All premises true.* Only sound arguments *prove* their conclusion.

Canonical illustration.

P1. All women are mortal.

P2. Socrates is a woman.

$\therefore$ **C.** Socrates is mortal.
This is **valid** (the form is correct) but **unsound** (P2 is false). Replace P1, P2 with the standard "All men are mortal; Socrates is a man"—now valid *and* sound. Note carefully: a valid argument can have all false premises and a true conclusion (e.g. "All heavenly bodies revolve around the Earth; the Sun is a heavenly body; therefore the Sun revolves around the Earth" is valid yet has a false conclusion—and it remains truth-preserving in the technical sense).

2.3.1 The Seven Combinations (Copi & Cohen, p. 30)

The truth-values of premises and conclusion can be permuted to yield seven non-trivial cases: valid+all-true, valid+all-false, invalid+all-true, invalid with true premises and false conclusion, valid with false premises and true conclusion, invalid with false premises and true conclusion, and invalid with all false propositions. Memorising the asymmetry—*a valid argument can never have all-true premises with a false conclusion*—is the heart of the matter.

2.4 The Laws of Thought

The Three Classical Laws of Logic

Law of Non-Contradiction
$\neg(A \wedge \neg A)$. Nothing can both possess and lack a property at the same time and in the same respect.
Law of Identity
$A = A$. Everything is identical to itself.
Law of the Excluded Middle
$A \vee \neg A$. For any property, everything either has it or lacks it.

Core Argument 3: Why Non-Contradiction Cannot Be Rationally Denied (Reductio Ad Absurdum)

P1. Suppose, for reductio, that the Law of Non-Contradiction is false: i.e. some proposition A is both true and false.

P2. To deny non-contradiction is itself to assert a definite claim ("it is the case that contradictions are admissible")—which *presupposes* that this very claim cannot also be its own negation, i.e. presupposes non-contradiction.

P3. Therefore the denier simultaneously affirms the principle.

$\therefore$ **C.** Non-contradiction is *presupposed by all rationality.* Without it, "we cannot believe things to be one way rather than another, and hence cannot think at all."

2.5 Logical vs. Physical Possibility

Logical / Physical Possibility & Impossibility

Physical impossibility
What is inconsistent with the laws of nature (science delimits this).

Logical impossibility
What is inconsistent with the laws of logic (e.g. *married bachelor*, *square circle*, *red and not-red at once*; all oxymorons).

Containment relation
Whatever is *physically* possible is *logically* possible—but not conversely. Archimedes's "*Give me a lever long enough... and I shall move the world*" is logically possible (no contradiction) but physically impossible.

2.6 Articulating Disagreement: The Russell–Coppleston Debate

The lecture cites the 1948 BBC debate between Russell (atheist/agnostic) and Coppleston (Catholic theologian) as the model of philosophical disagreement. Note its three opening moves:

1. **Definition.** Coppleston proposes: by "God" we mean a supreme personal Being, distinct from the world and its Creator. *Both sides accept this stipulation.*
2. **Position-articulation.** Coppleston: affirmative (God exists, and exists provably). Russell: *not* atheist (he does not claim non-existence is provable) but *agnostic.*
3. **Joint terrain.** Only after these clarifications does substantive disagreement begin.

Lesson. Genuine philosophical debate requires the *principle of charity* (§4)—a refusal to argue past each other. Compare to the modern television panel where "one mike is muted, the other amplified."

2.7 Hidden Premises

The Doctrine of Hidden Premises

Every inference we make in everyday life carries premises we never voice. "He must be guilty—he was arrested." The unstated premise: *police arrest only the guilty.* Logic's task: **press pause on pre-reflective judgment, surface the hidden premise, evaluate it.** The same applies to "two people seen together must be a couple," "the best-selling phone is the best phone," etc.

3 April 16 — Inference: Deduction, Induction, Conditionals

Thematic Overview

This session consolidated the deduction/induction distinction through worked classroom exercises before introducing *conditional* inference (modus ponens and modus tollens) alongside their fallacious mirrors. Three deeper theoretical threads ran through the latter half: Karl Popper's **falsificationism** as the deductive engine of science, Hume's **problem of induction** and its circularity, and the apparent paradox of why non-ampliative deduction is useful at all. The lecture closed with a preview of informal fallacies, bridging directly to the April 20 session.

3.1 Deduction vs. Induction: Worked Exercises

The class opened with a review of the distinction before working through exercises. The key diagnostic tests:

Diagnostic: Is an Argument Deductive or Inductive?

1. **Necessity test.** If the premises are true, *must* the conclusion be true? Yes $\Rightarrow$ (valid) deductive.
2. **Defeasibility test.** Can new evidence change the probability of the conclusion? Yes $\Rightarrow$ inductive.

Inductive arguments are never *valid/invalid*; they are *strong* or *weak*. Deductive arguments cannot be stronger or weaker—only valid or invalid.

Classroom Exercise Set

1. **Social media / progress.** "Those not on social media are anti-technology. Those who are anti-technology are anti-progress. Therefore those not on social media are anti-progress."
   Verdict: Valid deductive. *Sound?* No—the first premise is empirically suspect (e.g. Australia's social-media ban for under-16s does not entail anti-technology).
2. **Black cat.** "Each time a black cat crossed my path I got stuck in traffic. A black cat crossed my path this morning. Therefore I will be stuck in traffic."
   Verdict: Inductive—*weak*. Small sample; only a correlation, not a causal connection. This is the cognitive mechanism behind superstitions.
3. **Green tea.** "Green tea *contributes* to fat loss. I drink green tea. Therefore I will become thin."
   Verdict: Inductive—weak. "Contributes" is far weaker than "causes"; the conclusion claims more than the premises warrant.
4. **Exercise and type-2 diabetes.** "Exercise greatly reduces the risk of type-2 diabetes. I exercise regularly. Therefore my chances of getting type-2 diabetes are low."
   Verdict: Inductive—strong (given the premises). New information—family history, diet, insufficient exercise—can still alter the probability.
5. **All mammals / whales.** "All mammals have lungs. All whales are mammals. Therefore all whales have lungs."
   Verdict: Valid deductive, sound.
6. **Fort Knox conditional.** "If I owned all the gold in Fort Knox I would be wealthy. I do not own all the gold in Fort Knox. Therefore I am not wealthy."
   Verdict: Deductive, *invalid*—this is the fallacy of *denying the antecedent* (see §3.5).

3.2 Correlation vs. Causation

The black-cat and green-tea exercises prompted a classroom discussion on the limits of inductive inference.

Why Induction Yields Correlation, Not Causation

P1. Induction records *patterns*: A has followed B in all observed cases.

P2. To infer a *causal* connection goes beyond what observation strictly yields.

P3. The Scottish philosopher David Hume argued that all we ever observe are *constant conjunctions*—never the "necessary connection" we call causation.

$\therefore$ **C.** Science can establish high-probability correlations, but the step to causation requires additional philosophical machinery (counterfactual analysis, randomised controlled trials, mechanism).

Thinker Profile: David Hume (1711–1776)

Hume is the canonical sceptic about both causation and induction. On causation: we see the cue ball hit the nine ball but never see the *causing*—the mind imposes necessity where observation shows only sequence. On induction: no finite series of observations can logically guarantee a universal generalisation. Both scepticisms follow from his empiricist principle that all genuine knowledge derives from sensory experience. Immanuel Kant later replied that causation is a *category of the mind*—we impose it on experience—so its applicability to nature is guaranteed by the structure of cognition.

3.3 Conditionals: Necessary & Sufficient Conditions

Anatomy of "If A, then B"

Antecedent (A)
: The proposition following "if."

Consequent (B)
: The proposition following "then."

Sufficient condition
: A is *sufficient* for B: having A *guarantees* B.
: *Example.* Being a voter is sufficient for being over 18.

Necessary condition
: A is *necessary* for B: B cannot occur without A.
: *Example.* Oxygen is necessary for fire; it is not sufficient (fuel and heat also required).

Master rule
: In "If A, then B": the *antecedent* is a **sufficient** condition for the consequent; the *consequent* is a **necessary** condition for the antecedent.

3.4 Modus Ponens & Modus Tollens (Valid Forms)

Modus Ponens — Affirming the Antecedent

Form: P_1: If A, then B. P_2: A. $\therefore B$. (**Valid.**)

Classroom examples:

- "If it is raining, the ground will be wet. It is raining. $\therefore$ The ground is wet."
- "If you have a fever, you are sick. You have a fever. $\therefore$ You are sick."

Modus Tollens — Denying the Consequent

Form: P_1: If A, then B. P_2: $\neg B$. $\therefore \neg A$. (**Valid.**)

Classroom examples:

- "If my laptop is working, there is battery power. There is no battery power. $\therefore$ My laptop is not working."
- "If there is fire, there is oxygen. There is no oxygen. $\therefore$ There is no fire."

Note. A commonly confused version was also discussed: "If there is oxygen, there is fire. There is no fire. $\therefore$ There is no oxygen." This is *invalid* because oxygen is a necessary, not sufficient, condition for fire.

3.5 Conditional Fallacies (Invalid Mirrors)

Objection / Critique: Affirming the Consequent (Fallacy)

Form: If A, then B. B. $\therefore A$. **INVALID.**

Example. "If it is raining, the ground is wet. The ground is wet. $\therefore$ It is raining."
Error: The ground can be wet from sprinklers, flooding, etc. B does not exclude other sufficient conditions for B.

Objection / Critique: Denying the Antecedent (Fallacy)

Form: If A, then B. $\neg A$. $\therefore \neg B$. **INVALID.**

Example. "If I owned all the gold in Fort Knox, I would be wealthy. I do not own it. $\therefore$ I am not wealthy."
Error: Fort Knox gold is sufficient but not necessary for wealth; many other paths to wealth exist.

Example. "If you have a fever, you are sick. You do not have a fever. $\therefore$ You are not sick."
Error: You can be sick without having a fever.

Memorise: The Four Conditional Inferences

Move	**Name**	**Status**
$A \Rightarrow$ conclude B	Modus Ponens	**VALID**
$\neg B \Rightarrow$ conclude $\neg A$	Modus Tollens	**VALID**
$B \Rightarrow$ conclude A	Affirming the Consequent	**FALLACY**
$\neg A \Rightarrow$ conclude $\neg B$	Denying the Antecedent	**FALLACY**

3.6 Popper's Falsificationism: Science Uses Modus Tollens

The lecture argued that although science is often characterised as inductive, its core epistemic move is *deductive* (modus tollens).

Popperian Falsification as Modus Tollens

P_1 If theory T is true, then observational consequence O must obtain.

P_2 O does not obtain (experiment returns a contrary result).

$\therefore$ **C.** $\therefore$ Theory T is false.

Key insight. A single genuine counter-instance can *refute* a universal theory (modus tollens). No finite set of confirming instances can *prove* it (the problem of induction). This asymmetry is Popper's *falsificationism*.

Thinker Profile: Karl Popper (1902–1994)

Popper argued that the *demarcation criterion* separating science from pseudo-science is **falsifiability**: a claim is scientific iff it is in principle refutable by observation. Genuine science seeks to *disprove* hypotheses, not merely confirm them. The failure to design experiments that could falsify a hypothesis is a sign of **confirmation bias**—a tendency the lecturer linked to the replication crisis in contemporary psychology, where many studies designed only to confirm hypotheses fail to replicate.

Application: Cognitive Behavioural Therapy (CBT). The lecturer drew a vivid parallel: many psychological phobias are *implicit conditionals* ("If I take the lift, I will be trapped and die"). CBT's exposure therapy works by *falsifying* these conditionals—the patient takes the lift, nothing disastrous occurs, and the conditional is empirically refuted. Our behavioural repertoire is shaped by background conditional assumptions; therapy can be understood as applied Popperian reasoning. Similarly, the lecturer noted, much of our everyday reasoning is governed by implicit conditionals ("If I study AI I will get a job"; "If I follow my passion I will be unemployable")— bringing these to the foreground and evaluating them is itself a philosophical activity.

3.7 Hume's Problem of Induction

Objection / Critique: The Circularity of Inductive Justification

P_1 All inductive inference rests on the **Principle of the Uniformity of Nature** (PUN): the future will resemble the past.

P_2 The only available justification for PUN is that it has held reliably in the past—i.e. PUN is itself supported inductively.

P_3 Justifying induction by appeal to induction is **circular**.

$\therefore$ **C.** Inductive inference cannot be given a non-circular rational justification. Its warrant is ultimately *custom or habit*, not reason (Hume).

Responses discussed in class:

- *Pragmatic:* Induction *works*—it put humans on the moon (within Newtonian mechanics). A purely pragmatic vindication offers no logical foundation, but science does not need one to proceed within a research programme.
- *Kantian:* Causation is a *category of the mind*—we impose it on experience, so its applicability to nature is guaranteed by the structure of cognition (Kant reverses Hume's empiricism).
- *Status:* The problem remains open in philosophy of science. Scientists typically operate without worrying about foundations; philosophers continue to debate them.

3.8 Why Is Non-Ampliative Deduction Useful?

A student raised the challenge: if deduction adds no new information, why bother?

The Value of Non-Ampliative Inference

P1. The conclusion of a valid deductive argument is *already contained* in the premises—but it may not be *apparent* to a finite, cognitively limited reasoner.

P2. We are not purely rational animals; long chains of implication routinely surprise us.

$\therefore$ **C.** Deduction is testimony to human cognitive frailty: it makes *latent* truths *manifest*.

Cantor's infinities (classroom illustration). Consider $\mathbb{N} = \{1, 2, 3, \ldots\}$ and $\mathbb{E} = \{2, 4, 6, \ldots\}$. Intuitively $|\mathbb{N}| > |\mathbb{E}|$. Yet Cantor proved via a bijection $n \mapsto 2n$ that the two sets are the same size (both countably infinite). Furthermore, the set of all real numbers between 0 and 1 is *strictly larger* than $\mathbb{N}$—a result that violates intuition yet follows deductively from the definitions of sets and cardinality. The conclusion was always implicit in the axioms; we simply had not realised it. Proofs of this kind—still being produced in mathematics—justify deductive reasoning despite its non-ampliativity.

3.9 Preview: Formal vs. Informal Fallacies

The lecture closed by bridging forward to the April 20 session.

The Formal / Informal Distinction

Formal fallacies Errors of *structure*. The argument is invalid regardless of what A and B mean. Affirming the consequent is a formal fallacy.

Informal fallacies Errors of *content*. The argument may be structurally valid but the premises import irrelevant or manipulative material. Detection requires reading *meaning*, not just form. These go by the names: rhetoric, sophistry, demagogy.

Informal fallacies previewed in class (covered in full April 20):

- **Ad hominem:** attacking the arguer rather than the argument ("You're from the New York Times—terrible paper—so your question is invalid," cited as a Trump press-conference example).
- **Appeal to emotion:** invoking emotional responses instead of reasons ("Do this because I love you").
- **Ad baculum (appeal to force):** threats replace argument (international nuclear threats as the collapse of rational diplomacy).
- **Ad populum (appeal to popularity):** "Everyone believes aliens are real, therefore aliens are real." Majority opinion is not an independent support for truth.

***Thinker Profile: Plato on Democracy and the* Ad Populum**

Plato was sceptical of democracy partly because it institutionalises the *ad populum* fallacy: it grants truth-force to majority opinion. Socrates' execution (399 BCE) was cited as the paradigm case—the most famous victim of mob reasoning. All twelve thousand jurors voted guilty; their number did not make them right. The line between democracy and authoritarian majoritarianism, the lecturer noted, is philosophically thin and worth vigilance.

4 April 20 — Informal Fallacies: Fallacies of Relevance

Thematic Overview

The endsem core. Where §3 catalogued *formal* (structural) errors, this lecture turns to *informal* errors—arguments that look valid but whose content is flawed. The module's assigned readings (Copi & Cohen; Stich & Donaldson) organise informal fallacies into **four master categories**: (1) *Fallacies of Relevance*—premises are not connected to the conclusion; (2) *Fallacies of Defective Induction*—premises are related but too weak; (3) *Fallacies of Presumption*—an unproved assumption is smuggled in; (4) *Fallacies of Ambiguity*—a shift in word meaning or grammatical structure creates the illusion of support. Each category is treated below.

4.1 Sophistry vs. Fallacy — A Crucial Disambiguation

Fallacy / Sophistry

Fallacy
An argument that *appears* to support its conclusion but, on examination, does not. The term implies *nothing* about the speaker's motives—it may be deployed innocently. Hurley's formulation: "a courtroom trick."

Sophistry
The deliberate deployment of arguments *one knows* to be illogical or misleading. The motive is deception.

Baseline check
The film *Thank You for Smoking* (2005) is the recurring exemplar. Nick Naylor's line—"*If I prove you wrong, I don't have to prove I'm right*"—encapsulates the audience-capture function of fallacy in public debate.

4.2 The Four Master Categories of Informal Fallacy

1. **Fallacies of Relevance** (treated below).
2. **Fallacies of Defective Induction** (e.g. hasty generalisation, false cause).
3. **Fallacies of Presumption** (e.g. begging the question, complex question).
4. **Fallacies of Ambiguity** (e.g. equivocation, amphiboly).

4.3 Fallacies of Relevance: The Seven Forms

Defining feature. The premises offered have *no real connection* to the conclusion drawn. Because the connection is missing, the premises cannot establish the conclusion; because they are *made to appear* connected, the audience is deceived.

4.3.1 Appeal to the Populace (argumentum ad populum)

Fallacy: Appeal to Populace

"X is widely believed/done/bought; therefore X is true/right/best." The popularity of a proposition is irrelevant to its truth. **Examples (slides):** "This is the best-selling phone in India, so you should buy it." "A majority of the population thinks we coddle criminals; therefore it's time we got tough on crime." "UFOs must be real because a significant portion of the population takes UFO sightings seriously." **Bandwagon variant:** the urge to do what everyone else is doing. **Why bad:** the medieval populace held the Earth was the centre of the universe—numbers do not adjudicate truth.

4.3.2 Appeal to Emotion (ad misericordiam / ad passiones)

Fallacy: Appeal to Emotion

The argument substitutes for reasoning an appeal to emotions: pity, fear, patriotism, mercy, joy. "If you love your country, you will. . . " "If you drink Coca-Cola, your life will be wonderful."

Ad misericordiam (appeal to pity) is especially common in legal contexts: a defence attorney describes a client's terrible childhood to win acquittal, when the question at issue is simply whether the client committed the crime. Pity is relevant to *sentencing*, but irrelevant to the question of *guilt*.

Crucially: emotions are not *always* irrelevant—a mercy-petition properly considered may be morally weighty—but they become *fallacious* when they replace, rather than supplement, reasoned grounds for the conclusion.

4.3.3 Red Herring

Fallacy: Red Herring

"The arguer *diverts the attention* of the reader or listener by *changing the subject* to a different but sometimes subtly related one, and then either draws a conclusion about this different issue or merely presumes that some conclusion has been established." **Slide example.** "Environmentalists are continually harping about the dangers of nuclear power. Unfortunately, electricity is dangerous no matter where it comes from. Every year hundreds of people are electrocuted . . . " The original topic (nuclear power's distinctive risks) is silently swapped for "electricity in general."
Sub-form (whataboutism). "You're arguing X? What about Y?"

4.3.4 Straw Man

Fallacy: Straw Man

The arguer **distorts** the opponent's position into a weakened or exaggerated version that is easier to attack, then attacks the distortion. **Tell-tale phrases:** "So what you're saying is. . . "; "the real reason. . . " **Class example.** Opponent: "AI may be conscious." Straw-man reply: "So you think chairs can be conscious too? Should we give chairs rights?" The opponent's view (machines may instantiate consciousness) is replaced by an absurd surrogate (anything material is conscious). **Antidote: the Principle of Charity.** *Reconstruct the opponent's argument in its strongest plausible form before refuting it.*

4.3.5 Ad Hominem (Attack on the Person)

The most common and most variously sub-typed fallacy of relevance. Its forms:

Kinds of Ad Hominem

Abusive
: Attacks the speaker's character. "Maher is an arrogant, shameless pig; therefore his arguments are not worth listening to." Includes character assassination and **guilt by association** ("don't listen to her—she hangs out with so-and-so").

Poisoning the well
Pre-emptively attacks the *motivations* underlying an argument: "Of course philosophers will argue against AI—it threatens their profession." "Kejriwal talks about the economy because it benefits him politically." The argument is rejected without engaging its content.

Circumstantial
Points to a perceived conflict of interest in the speaker's circumstances rather than addressing the argument.

Tu quoque ("you too")
Rebuts an argument by accusing the arguer of the same fault. "Don't tell me to quit smoking—you smoke." "How can you advise me on my marriage? You're divorced." The speaker's hypocrisy, even if real, is irrelevant to the truth of the claim.

When attacks on character *are* relevant. If the matter under debate is precisely whether the speaker is qualified to act in some role (e.g. a witness's truthfulness, a candidate's integrity), then character becomes a legitimate premise. The fallacy lies in treating character as a substitute for engaging with the argument's content.

4.3.6 Appeal to Force (ad baculum)

Fallacy: Appeal to Force

Persuasion by threat. "Believe my conclusion or face consequences." The threat is causally efficacious in producing assent, but provides no *rational* ground for accepting the conclusion.

4.3.7 Missing the Point (ignoratio elenchi)

Fallacy: Missing the Point

The premises support *some* conclusion—but *not* the one drawn. The arguer leaps from premise to a vaguely related but unwarranted conclusion. **Slide examples.**

- "Crimes of theft and robbery have been increasing alarmingly. Therefore we must reinstate the death penalty." (The premise might support *some* response—better policing, addressing unemployment—but not the leap to capital punishment.)
- "Abuse of the welfare system is rampant. Our only alternative is to abolish the system altogether."

Distinguishing principle (slide caveat). *Both red herring and straw man proceed by introducing a new set of premises. Missing the point does not—there is only a leap from the same premise to a wrong conclusion.* Reserve the name *ignoratio elenchi* for fallacies of irrelevance *that do not fit any other category.*

Non sequitur ("it does not follow") is the informal label applied when the gap between premises and conclusion is so glaringly obvious that no specific fallacy name is needed. It subsumes any argument whose conclusion simply fails to follow.

4.4 Disambiguating Red Herring vs. Straw Man vs. Missing the Point

This is the lecture's most pedagogically emphasised distinction (instructor's warning: *do not give multiple labels for one fallacy in the exam*).

Triadic Comparison Chart

Straw man
The opponent's premise A is replaced by a weakened version A^-; the arguer attacks A^- and claims to have defeated A.

Red herring
The opponent's premise A is *ignored*; an unrelated premise B is introduced and a conclusion is drawn from B.

Missing the point
No new premise is introduced; the original premise A supports some conclusion C_1, but the arguer *leaps* to a different, unwarranted conclusion C_2.

Test. *Did the arguer knock down a distorted argument (straw man) or simply change the subject (red herring)?* If neither, but the leap is from premise to wrong conclusion: *missing the point.*

4.5 Fallacies of Defective Induction (D1–D4)

In these fallacies the premises are not wholly irrelevant to the conclusion— they are genuinely related—but they are far too weak or insufficient to support it.

4.5.1 Argument from Ignorance (ad ignorantiam)

D1: Argument from Ignorance

Arguing that something is true because it has not been proved false, or false because it has not been proved true.

Examples. "Nobody has disproved that UFOs are extraterrestrial; therefore they are." "We cannot prove this new drug is harmful; therefore it is safe."

Principle: Absence of evidence is not always evidence of absence.

Exception: In rigorous scientific testing (e.g. drug trials), the *systematic* failure to detect harmful effects after thorough investigation can constitute evidence of safety—but this is entirely different from merely asserting that no one has disproved a claim.

4.5.2 Appeal to Inappropriate Authority (ad verecundiam)

D2: Appeal to Inappropriate Authority

Appealing to an authority who has no genuine expertise in the relevant field, or citing authority in a domain where experts legitimately disagree.

Examples. A famous actor endorsing a medication. A physicist giving opinions on economic policy. A celebrity advocating a dietary supplement.

Key test: Is this person genuinely expert in the *specific matter at hand?* Appealing to genuine experts in their own domain is not fallacious—it becomes so only when expertise is absent or the domain is one of legitimate expert disagreement.

4.5.3 *False Cause (non causa pro causa)*

D3: False Cause — *Post Hoc* and Slippery Slope

Post hoc ergo propter hoc ("after this, therefore because of this"): because B followed A, it is concluded that A caused B.

Examples.

- "The sun rises every morning after people sleep; therefore sleep causes the sunrise."
- "Each time a black cat crossed my path, I got stuck in traffic." (Correlation mistaken for causation—the superstition mechanism.)

Slippery slope variant: Arguing that one step in a direction will *necessarily* lead to a cascade of increasingly bad outcomes, without demonstrating each causal link.
Example. "If we allow assisted suicide, doctors will soon be killing patients without consent."
Error: The feared cascade is merely asserted; each step would require its own separate argument.

4.5.4 *Hasty Generalisation (converse accident)*

D4: Hasty Generalisation

Drawing a broad general conclusion from too few or unrepresentative cases. Stereotypes are the most common instance: judging all members of a group by one or two encounters.

Classic example. "Every swan I have ever seen is white; therefore all swans are white." This was a plausible inductive inference for European naturalists—until black swans were discovered in Australia. One counter-instance defeated the generalisation. To generalise responsibly, the sample must be sufficiently large *and* representative of the target population.

4.6 Fallacies of Presumption (P1–P3)

These fallacies smuggle in an unwarranted assumption that the argument depends upon but never proves.

4.6.1 *Accident*

P1: Accident

The reverse of hasty generalisation. Instead of moving incorrectly from a few cases to a general rule, *accident* involves incorrectly applying a general rule to a particular case where it does not apply because of special ("accidental") circumstances. Almost every rule has exceptions; ignoring them commits this fallacy.

Example. "You should always keep your promises" is a good general rule, but it does not apply if keeping the promise would require harming someone. The rule is applied blindly to a case that falls outside its intended scope.

4.6.2 *Complex Question (plurium interrogationum)*

P2: Complex Question

Asking a question in a way that presupposes the truth of a conclusion buried within it. Any answer appears to confirm the hidden assumption.

Classic example. "Have you stopped beating your wife?" Both "yes" and "no" imply that at some point you were beating her.

Contemporary example. “With all the hysteria and phony science, isn’t man-made global warming the greatest hoax ever?” This presupposes the science is phony—a contested claim that should be argued for, not buried in a question.

Antidote: Refuse to answer on the question’s own terms; expose the embedded assumption and challenge it directly.

4.6.3 Begging the Question (petitio principii)

P3: Begging the Question (Circular Argument)

Also called **circular reasoning**. The conclusion of the argument is hidden in one of the premises, often disguised by different wording. The argument goes in a circle, proving nothing that was not already assumed.

Classic example. “Freedom of speech must be advantageous to the state, because it is conducive to the community’s interests for individuals to enjoy unlimited freedom of expression.”
Analysis: The conclusion (freedom of speech is good for the state) is restated as the premise (unlimited expression is conducive to community interests). No independent support is offered.

Important note. Circular arguments are always technically *valid*—the conclusion does follow from the premise—but they are *worthless*: they prove nothing that was not already assumed. This is why *soundness*, not just validity, is the ultimate standard.

4.7 Fallacies of Ambiguity (A1–A5)

These fallacies arise because words or phrases shift meaning within the argument, or because the grammatical structure is ambiguous.

4.7.1 Equivocation

A1: Equivocation

Using the same word in two *different senses* within the same argument. The shift in meaning is often subtle.

Examples.
- “The sign said ‘fine for parking here,’ so I parked.” (“fine” = penalty vs. “fine” = acceptable.)
- “All laws have a lawmaker. Nature has laws. Therefore nature has a lawmaker.” (“laws” = legal statutes vs. scientific regularities.)

Note: Equivocation is especially common with *relative terms* (“tall,” “good,” “small”) whose meaning shifts with context. A “good general” and a “good president” require very different kinds of goodness.

4.7.2 Amphiboly

A2: Amphiboly

The fallacy arises when the *grammatical construction* of a sentence makes it ambiguous—two interpretations are possible—and the argument uses one interpretation as a premise but draws a conclusion that only follows from the other.

Example. “Women prefer Democrats to men.” Does this mean women prefer Democrats over men, or that women prefer Democrats to Republicans?

4.7.3 Accent

A3: Accent

Meaning is distorted by misleading emphasis on certain words, or by taking words out of context so that a different meaning is conveyed.

Example. A theatre critic writes "it is *far from* the funniest play on Broadway"; an advertiser quotes her as saying "funniest play on Broadway!"

Common contexts: Political propaganda, advertising (large-print offers with "and up" in tiny print), and out-of-context quotation.

4.7.4 Composition

A4: Composition

Reasoning incorrectly from what is true of the *parts* to what must be true of the *whole*.

Examples.

- Each part of a machine is light in weight; therefore the whole machine is light.
- Each individual musician in an orchestra is skilled; therefore the orchestra will perform brilliantly.

Error: The whole can have emergent properties different from those of its parts.

4.7.5 Division

A5: Division

The reverse of composition: reasoning incorrectly from what is true of the *whole* to what must be true of each *part*.

Examples.

- The orchestra is world-famous; therefore every individual musician in it is world-famous.
- The university is excellent; therefore every department in it is excellent.

Error: What is true collectively of a group is not necessarily true of each of its individual members.

4.8 The Principle of Charity & the Note on Refutation

Methodological Principles for Honest Disagreement

Principle of Charity. When evaluating an opponent's argument, *interpret it in the strongest possible form* consistent with their words. To attack a weaker version is to commit straw man.

Note on Refutation. Refuting one bad argument for C does *not* establish $\neg C$. Every true proposition could be the conclusion of a terrible argument: "All men are mortal; Thursday follows Wednesday; *therefore* New Delhi is the capital of India." The conclusion is true; the argument is awful. Hence: **rational reconstruction**—steel-man your opponent's case before you attack it.

4.9 How to Discover a Fallacy — Procedure

1. Identify the **conclusion**—what is the argument trying to establish?

2. Identify the **premises**—what reasons are being offered?
3. Ask: do the premises *independently and genuinely* support the conclusion?
4. If **no**, the argument is fallacious. Ask *which kind*:
   - Are the premises *completely irrelevant*? → *Fallacy of Relevance* (ad hominem, appeal to emotion, red herring, straw man, ad baculum, ad populum, missing the point).
   - Are the premises relevant but *too weak*? → *Fallacy of Defective Induction* (ad ignorantiam, ad verecundiam, false cause / post hoc, hasty generalisation, slippery slope).
   - Is a *hidden false assumption* smuggled in? → *Fallacy of Presumption* (accident, complex question, begging the question).
   - Is a *word or phrase shifting meaning*? → *Fallacy of Ambiguity* (equivocation, amphiboly, accent, composition, division).
5. Within the appropriate category, name the specific fallacy:
   - Are the premises attacking the person, character, or motivation? → *Ad hominem* variants.
   - Are they appealing to emotion, popularity, or force? → *Fallacies of relevance.*
   - Are they based on too few cases? → *Hasty generalisation.*
   - Are they restating the conclusion as a premise? → *Begging the question.*
   - Is a false or misleading causal connection assumed? → *False cause* (post hoc).
   - Is the conclusion different from what the premises actually establish? → *Missing the point / non sequitur.*

One-label rule (exam etiquette). Give *one* principal fallacy label per argument. Do not hedge with multiple names. Exam items are constructed to be unambiguous instances of a single dominant fallacy.

4.10 Worked Exam-Style Diagnoses

Diagnostic Walkthrough I (Slide example)

"Mr. Goldberg has argued against prayer in the public schools. Obviously Mr. Goldberg advocates atheism. But atheism is what they used to have in Russia. Atheism leads to the suppression of all religions and the replacement of God by an omnipotent state. Is that what we want for this country? Clearly Mr. Goldberg's argument is nonsense."

Diagnosis. The argument leaves Goldberg's actual claim (against school prayer) unaddressed and instead constructs a weak surrogate ("Goldberg endorses Soviet-style atheism"). This is a **straw man**— the original premise is replaced by a distorted, easily-refuted version.

Diagnostic Walkthrough II (Slide example)

"Your gun control proposal is an affront to gun owner rights. We are talking about liberty being seized by an overinvasive government here." — "Oh, that's funny coming from the person who proposed such draconian abortion regulations."

Diagnosis. The reply does not engage the gun-control argument; it points to the speaker's own past inconsistency. **Tu quoque** (circumstantial ad hominem). One might secondarily note a *red herring* shift to abortion—but per the instructor's caveat, give a *single* label that captures the dominant move.

Diagnostic Walkthrough III (Slide example)

"The United States says North Korea is stockpiling nuclear weapons and is therefore a danger to world peace. But surely both Israel and the United States have their own stockpiles!"

Diagnosis. Tu quoque. The accusation may be true, but it does not engage with whether North Korean stockpiling is in fact dangerous.

Diagnostic Walkthrough IV (Slide example)

"You say caste-based reservation is a way to correct historical wrongdoings. Then what about the economically poor who are from upper castes? Should we just ignore them?"

Diagnosis. Red herring. The original premise concerns correction of caste-based historical injustice. The reply diverts to a distinct (and not mutually exclusive) issue—economic poverty. *The two are not incompatible*: addressing one does not preclude addressing the other.

4.11 Why We Fall for Fallacies (Kahneman's System 1 / System 2)

Why Fallacies Persuade Despite 3,000 Years of Catalogue

P1. Humans are not purely rational; we are biological beings with urges, drives, and emotional susceptibilities.

P2. Per Kahneman (*Thinking, Fast and Slow*), cognition operates in two modes: **System 1** (fast, automatic, intuitive) and **System 2** (slow, deliberate, effortful).

P3. Fallacies are crafted to engage System 1—to bypass deliberation by appealing to emotion, identity, popularity, fear.

∴ **C.** Logic's normative work is to interrupt the System 1 default and *engage System 2*—"press pause, evaluate the inference, surface the hidden premise." This is what philosophy as "*the region of liberating doubt*" practically means.

5 Master Lexicon: Philosophy & Logic Terms

Abductive inference
Inference to the best explanation.

Accent Fallacy of ambiguity: meaning distorted by misleading emphasis or by taking words out of context. (See §4.7.)

Accident
Fallacy of presumption: incorrectly applying a general rule to a particular case where special circumstances make it inapplicable.

Ad baculum
Fallacy of appeal to force.

Ad hominem
Fallacy of attacking the person rather than the argument.

Ad ignorantiam
Fallacy: arguing that something is true because it has not been proved false, or false because not proved true. Absence of evidence is not always evidence of absence.

Ad misericordiam
Fallacy of appeal to pity; sub-type of appeal to emotion. Especially common in legal settings (describing a defendant's troubled past to win acquittal on a guilt question).

Ad populum
Fallacy of appeal to popularity (bandwagon fallacy).

Ad verecundiam
Fallacy of appeal to inappropriate authority: citing someone as authoritative in a domain outside their expertise, or where experts legitimately disagree.

Affirming the antecedent
Valid form: modus ponens.

Affirming the consequent
Fallacy mirroring modus ponens.

Amphiboly
Fallacy of ambiguity arising from grammatical construction: the sentence can be interpreted two ways; the argument uses one interpretation as premise but draws a conclusion that follows only from the other.

Ampliative
Of an inference: providing genuinely new information beyond what the premises strictly contain. (Inductive arguments are ampliative; deductive are not.)

Antecedent
In "If A, then B," the proposition A.

Argument
A structure in which premises are offered in rational support of a conclusion.

Axiology
: Theory of value.

Begging the question (*petitio principii*)
: Fallacy of presumption: the conclusion is hidden in one of the premises (circular argument). Always technically valid but worthless as proof.

Categorical proposition
: A proposition of the four classical Aristotelian forms: All S are P; No S are P; Some S are P; Some S are not P.

Conclusion
: The proposition an argument seeks to establish.

Complex question (*plurium interrogationum*)
: Fallacy of presumption: a question that presupposes the truth of a conclusion buried within it, so any answer confirms the assumption. ("Have you stopped beating your wife?")

Composition
: Fallacy of ambiguity: reasoning incorrectly from what is true of the parts to what must be true of the whole.

Consequent
: In "If A, then B," the proposition B.

Deductive inference
: Inference whose conclusion follows from its premises with absolute necessity.

Denying the antecedent
: Fallacy mirroring modus tollens.

Denying the consequent
: Valid form: modus tollens.

Descriptive
: Concerning what *is*; the domain of empirical science.

Division
: Fallacy of ambiguity: reasoning incorrectly from what is true of the whole to what must be true of each part.

Dunning–Kruger effect
: Cognitive bias whereby the less one knows of a domain, the more one overestimates one's competence.

Epistemology
: Theory of knowledge.

Equivocation
: Fallacy of shifting a word's meaning across premises.

Fact–value dichotomy
: The thesis that descriptive facts cannot, alone, entail normative values.

Fallacy
An argument that appears valid but, on examination, is not.

Hasty generalisation (*converse accident*)
Fallacy of defective induction: drawing a broad conclusion from too few or unrepresentative cases. The black-swans example is canonical.

Hidden premise
An unstated proposition without which the inference does not go through.

Hypothetico-deductive method
Popper's account of scientific method: conjecture a theory, derive testable predictions, attempt to falsify.

Ignoratio elenchi
"Missing the point"; the catch-all fallacy of irrelevance.

Inductive inference
Inference whose conclusion is supported by premises only with probability.

Is–ought gap
The Humean thesis that no normative conclusion follows from purely descriptive premises.

Law of excluded middle
$A \lor \neg A$.

Law of identity
$A = A$.

Law of non-contradiction
$\neg(A \land \neg A)$.

Liar's paradox
"This sentence is false." Self-referential paradox that motivates type theory.

Logic The normative discipline of right reasoning.

Logical impossibility
Inconsistency with the laws of logic (e.g. square circle).

Metaphysics
Inquiry into the most general structure of reality.

Modus ponens
Valid form: $A \to B,\ A,\ \therefore B$.

Modus tollens
Valid form: $A \to B,\ \neg B,\ \therefore \neg A$.

Necessary condition
A is necessary for B iff $\neg A \to \neg B$.

Non-monotonic
Of inference: the addition of new premises may overturn the conclusion (inductive

arguments are non-monotonic).

Non sequitur
: "It does not follow": informal label for any argument in which the gap between premises and conclusion is glaringly obvious.

Normative
: Concerning what *ought* to be; the domain of philosophy.

Ontology
: Theory of being / what exists.

Petitio principii
: See *Begging the question.*

Physical impossibility
: Inconsistency with the laws of nature.

Poisoning the well
: Sub-type of ad hominem: pre-emptive attack on motives.

Post hoc ergo propter hoc
: "After this, therefore because of this": false-cause fallacy inferring causation from temporal sequence.

Premise
: A proposition advanced as part of an argument's support.

Principle of charity
: Methodological principle: interpret the opponent's argument in its strongest plausible form.

Principle of the uniformity of nature
: The presupposition of all inductive inference: future regularities will resemble past.

Proposition
: The truth-bearing meaning expressed by a declarative sentence.

Reductio ad absurdum
: Method of disproving A by deriving a contradiction from A (used to defend non-contradiction).

Sentence
: A linguistic entity (in contrast to the proposition it expresses).

Slippery slope
: Variant of false-cause fallacy: asserting that one action will inevitably lead to a chain of bad consequences, without demonstrating each causal link.

Socratic irony
: The acknowledged recognition of one's own ignorance.

Sophistry
: Deliberate use of arguments one knows to be misleading.

Soundness
: An argument is sound iff it is valid *and* all premises are true.

Straw man
: Fallacy of attacking a distorted version of the opponent's argument.

Sufficient condition
: A is sufficient for B iff $A \to B$.

Syllogism
: Aristotelian three-line argument with two premises and a conclusion.

Truth A property of propositions; correspondence to the way the world is.

Truth-preserving
: Of inference: when valid, the conclusion cannot be false if the premises are true.

Tu quoque
: Sub-type of ad hominem: "you too."

Validity
: A property of arguments; the impossibility of true premises with a false conclusion.

Whataboutism
: Colloquial label for the red-herring tactic.

6 Endsem Practice: Argument-Construction & Fallacy Diagnosis

6.1 Fallacy-Identification Drill

For each of the following (drawn from class transcripts and slides), identify *the principal* fallacy and explain in one sentence why it fits.

1. "Look at the line for that new movie! It is out of the theatre door, down the street, and around the corner. That film must be great."

2. "There's no point going to him for advice on your marriage. He's a divorcee."

3. "Philosophers will obviously argue against the possibility of AI, since it puts their whole profession at the risk of being replaced."

4. "Many children have developed autism shortly after having been given the MMR jab; therefore the MMR jab causes autism."

5. "Since the 1950s, atmospheric CO_2 levels and crime levels have both increased sharply; therefore atmospheric CO_2 causes crime."

6. "Those who resist affirmative-action programmes deny that they are racists, but the truth is that their real motivation is racism."

7. "Each time a black cat crossed my path, I got stuck in a traffic jam; a black cat crossed my path this morning, therefore I will be stuck in a traffic jam."

Suggested answers (justify in your own words): (1) ad populum; (2) circumstantial ad hominem; (3) poisoning the well (ad hominem); (4) false cause / *post hoc ergo propter hoc*; (5)

false cause / spurious correlation; (6) poisoning the well; (7) hasty generalisation (small, possibly unrepresentative sample; correlation only).

Extended drill — new categories:

8. "Nobody has proved that ghosts don't exist; therefore ghosts exist."
9. "The university is the best in the country; therefore every course taught there is excellent."
10. "The sign said 'fine for parking here,' so I parked."
11. "All laws have a lawmaker; nature has laws; therefore nature has a lawmaker."
12. "Have you stopped cheating in exams?"
13. "Freedom of speech is good because it is conducive to free expression."
14. "If we allow euthanasia, doctors will soon be killing patients without consent."
15. "This celebrity endorses this supplement, so it must work."
16. "You should always keep your promises—give back the knife you borrowed from your friend who is now in a murderous rage."

Suggested answers (8–16): (8) ad ignorantiam; (9) division; (10) equivocation ("fine"); (11) equivocation ("laws"); (12) complex question; (13) begging the question / petitio principii; (14) slippery slope; (15) ad verecundiam; (16) accident.

6.2 Conditional Fallacy Drill

For each, identify whether the inference is *modus ponens*, *modus tollens*, *affirming the consequent*, or *denying the antecedent*.

1. "If there's oxygen, then there's fire. There's no fire; therefore there's no oxygen." (*modus tollens form—but the conditional itself is false: oxygen is necessary, not sufficient, for fire*)
2. "If there's fire, then there's oxygen. There's fire; therefore there's oxygen." (*modus ponens; valid*)
3. "If there's fire, then there's oxygen. There's no oxygen; therefore there's no fire." (*modus tollens; valid*)
4. "If I owned all the gold in Fort Knox, I would be wealthy. I do not own all the gold in Fort Knox; therefore I am not wealthy." (*denying the antecedent; fallacy*)
5. "If this liquid is acidic, it will turn litmus paper blue. This liquid does not turn litmus paper blue; therefore this liquid is not acidic." (*modus tollens; valid*)

6.3 Argument-Reconstruction Exercises

1. Reconstruct, in numbered-premise form, Russell's argument (April 9 lecture) that the value of philosophy lies not in delivering certainty but in cultivating openness.
2. Reconstruct Searle's Chinese Room argument (discussed in §1 under "Philosophy as the Root of Other Disciplines"): state the four axioms formally and identify which axiom a defender of strong AI would most plausibly deny.

3. Reconstruct the argument behind the slogan "*We need philosophy because bad philosophy is all around*" (Putnam): articulate the implicit premises that connect "bad philosophy is around" to "we ought to do good philosophy."

6.4 Concept-Distinction Essay Prompts

1. Distinguish *validity* from *soundness*, illustrating with an argument that is valid but unsound, and explain why deductive validity does not, by itself, guarantee that the conclusion is true.

2. Distinguish *deductive* from *inductive* inference along the four axes discussed (necessity vs. probability; non-ampliative vs. ampliative; monotonic vs. non-monotonic; *a priori* vs. *a posteriori* appraisal).

3. Distinguish *red herring* from *straw man* from *missing the point*; provide one original example of each.

4. Explain the *is–ought gap* and assess whether scientific evidence about, say, deterrent effects of capital punishment, can by itself settle whether capital punishment ought to be implemented.

5. Explain the *Principle of the Uniformity of Nature* and the problem it raises for the rational justification of inductive inference.

6.5 Final Examiner's Tip Sheet

Instructor's Stated Exam-Etiquette

- Material is examinable from **slides, readings, AND classroom discussion**—not just slides.
- For fallacy-identification questions: give **one** unique label; do *not* hedge with multiple names. Exam items will be unambiguously instances of a single principal fallacy.
- Reserve *ignoratio elenchi* ("missing the point") for fallacies of irrelevance *that do not fit any other category.*
- Distinguish red-herring vs. straw-man by asking whether the arguer *generated a new premise* (red herring / straw man) or merely *leapt to a wrong conclusion from the same premise* (missing the point).

End of synthesis. "Philosophy, though unable to tell us with certainty what is the true answer to the doubts which it raises, is able to suggest many possibilities which enlarge our thoughts and free them from the tyranny of custom." — Bertrand Russell, *The Problems of Philosophy.*